

Paper.id: Improving cash flow for local SMEs on a secure cloud platform

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[About Paper.id](#)

Paper.id is easing
flow burden of SMEs in
Indonesia with its
and scalable pay
platform hosted on
Cloud.



About Paper.id

Paper.id was founded in 2017 to help small and medium enterprises (SMEs) in Indonesia digitize their invoicing and payment systems, so they can better manage their cash flow. Paper.id provides various features for businesses such as document tracking and matching, automatic payment reconciliation, and helps ease supply chain financing by invoicing in one platform. As a leading B2B Software-as-a-service (SaaS) invoicing startup in Indonesia, Paper.id has raised \$12 million in Series B funding.

Google Cloud results

- Ensures
99.99%
uptime with

Accel
mark

- Meets ISO 27001 certification and other regulatory compliance requirements
- Reduces regression test time from 17 hours to just 1.5 hours
- Encourages faster time-to-market and shorter sprint periods

Location: Indonesia

 Cloud Ace

Cloud Ace is a managed service provider for Google Cloud with 100 engineers in Indonesia, Japan, Singapore, Taiwan, Thailand, and Vietnam. It provides one-stop services such as cloud implementation support, operational design, and post-implementation system maintenance so that customers get the most out of their cloud solutions.

VALUATION

ABOUT Cloud Ace

Cloud Composer

Cloud Storage (<https://cloud.google.com/storage/>)

(<https://cloud.google.com/kubernetes-engine/>)

(<https://cloud.google.com/products/operations>)

For most micro, small and medium enterprises (MSMEs), ensuring a steady cash flow to keep the business running smoothly is crucial because business revenue from receivables or accounts payable flow is crucial because businesses need to ensure they get paid on time from customers, so that they can maintain their suppliers and employees.

A full-stack B2B payment platform at the end of 2016, Paper.id (<https://paper.id/>) supports and boosts MSMEs so suppliers get paid faster. Meanwhile, the flexibility to pay on their own terms with credit cards. With more than 470,000 users across different business segments, the company can be integrated with large enterprise ERP systems via API, or act as an end-to-end solution for MSMEs to connect and digitize their entire supply chain process.

Logging (<https://cloud.google.com/logging/>)

most out of their cloud solutions

Monitoring (<https://cloud.google.com/monitoring/>)

As a payment gateway that offers more than 30 different payment methods including bank transfer, QR payments, credit card payments, and even options to pay through various marketplaces, security is always top of mind. "We're always on the lookout for possible data breaches, unauthorized access, and other security threats, as our customers' businesses are at stake," says Jeremy Limman, co-founder and Chief Product Officer at Paper.id.

With the support of Cloud Ace (<https://id.cloud-ace.com/>), the company has been with Google Cloud (<https://cloud.google.com/>) for its production stack for three years, having moved from a smaller-scale platform. "We launched on a much simpler platform that was also fairly easy to set up. However, as we grew as a company, we needed a solution that could support the traffic coming through, as well as something that would enable us to scale. We found Google Cloud to be the ideal solution for that," says Mr. Jeremy.

Since implementing Google Cloud, Jeremy and his team have received support from Cloud Ace. "In the initial stages, Cloud Ace recommended some products like BigQuery

(<https://cloud.google.com/bigquery>) for us to get familiar with the platform, and since then, they've been working alongside us in providing advice on the best products that would meet our business needs as we grow," shares Jeremy. Cloud Ace also conducts training sessions which are tailored to Paper.id's needs. Being in Indonesia, Jeremy shares that being able to have engineers communicate in Indonesian further builds rapport and an easy working relationship.

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Fulfilling regulatory compliance requirements

As a financial solution platform, there is a host of regulatory compliance that needs to be met. One of them is ensuring that all data resides in the country. "We migrated in 2020, around the time that Google Cloud launched its Indonesia Cloud Region. This was crucial for us in our decision to come on board since regulatory compliance was a key component of our decision-making process in choosing a cloud provider," says Jeremy.

As an ISO 27001-certified company, Paper.id places a high value on the various security certifications that Google Cloud has been awarded with. "As we add more services into our infrastructure, we need to ensure that every new component comes with the

corresponding security measures like identity and access management (IAM), for example, which our previous cloud provider did not have," adds Jeremy.



Platform simplicity for ease of use

As a business in a fast-moving environment, Paper.id values a cloud platform that's simple and easy to use. "The simplicity of the Google Cloud dashboard under the Operations Suite (<https://cloud.google.com/products/operations>) makes it easy to navigate through the different tools and monitor the backend," says Jeremy. "You don't have to go through multiple courses just to use the dashboard, and that in itself has been a plus point for us." Using the alerts feature on Cloud Logging (<https://cloud.google.com/logging>) and Cloud Monitoring (<https://cloud.google.com/monitoring>), Paper.id is able to support a 99.99 percent uptime.

With just three DevOps engineers, Paper.id is able to manage and maintain its entire infrastructure, leveraging Google Kubernetes Engine (<https://cloud.google.com/gke>) (GKE), paired with highly available clusters and node pools to serve 100 percent service hosting, monitoring, and scaling. "Google Kubernetes Engine gives us the flexibility to scale up and down depending on traffic, which has been really helpful in terms of managing both cost and availability," says Jeremy.

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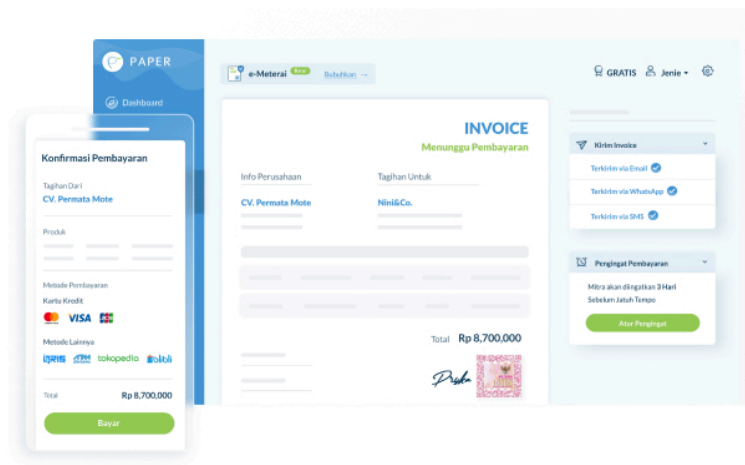
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Serving millions of invoices quickly and efficiently

Since coming onboard Google Cloud, Paper.id achieved an average application index (Apdex) score of 0.93 within its target of 500ms, with a minimal error rate of just 0.6 percent. At the same time, DevOps has been able to reduce regression test time from 17 hours to just 1.5 hours for new product launches, with productivity at lead time having improved from a peak of 28 days to 11 days, below one 14-day sprint period.

With millions of invoices going through its platform every week, Paper.id relies heavily on Cloud Storage (<https://cloud.google.com/storage>) for document storage as well as its data lake, alongside Cloud Composer (<https://cloud.google.com/composer>) to store and effectively manage customer documents within a secure environment.



"We've actually begun exploring Vertex AI (<https://cloud.google.com/vertex-ai>) in a few different ways since 2022 and have done some piloting and experimentation," shares Jeremy. "We're now looking into embedding Vertex AI into some of our workflows to leverage the machine learning capabilities."

Jeremy explains that Vertex AI would be used to support reading and processing documents like invoices, receipts, purchase orders, agreements, even full financial statements. "There are a lot of documents in the B2B space, and I believe what Vertex AI helps us do is to remove a lot of the manual work. I can say with certainty that this has accelerated our growth in the market by three times," says Jeremy.

Staying true to its mission of supporting local SMEs in Indonesia, Jeremy shares, "We are only just scratching the surface of the possibilities for B2B payments and SMEs in Indonesia. There are many products in our pipeline, such as a proprietary fraud detection engine, specialized AI for businesses and biometric verification, and with Google Cloud, we

want to help these companies do their invoicing more seamlessly as they connect with enterprises."

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